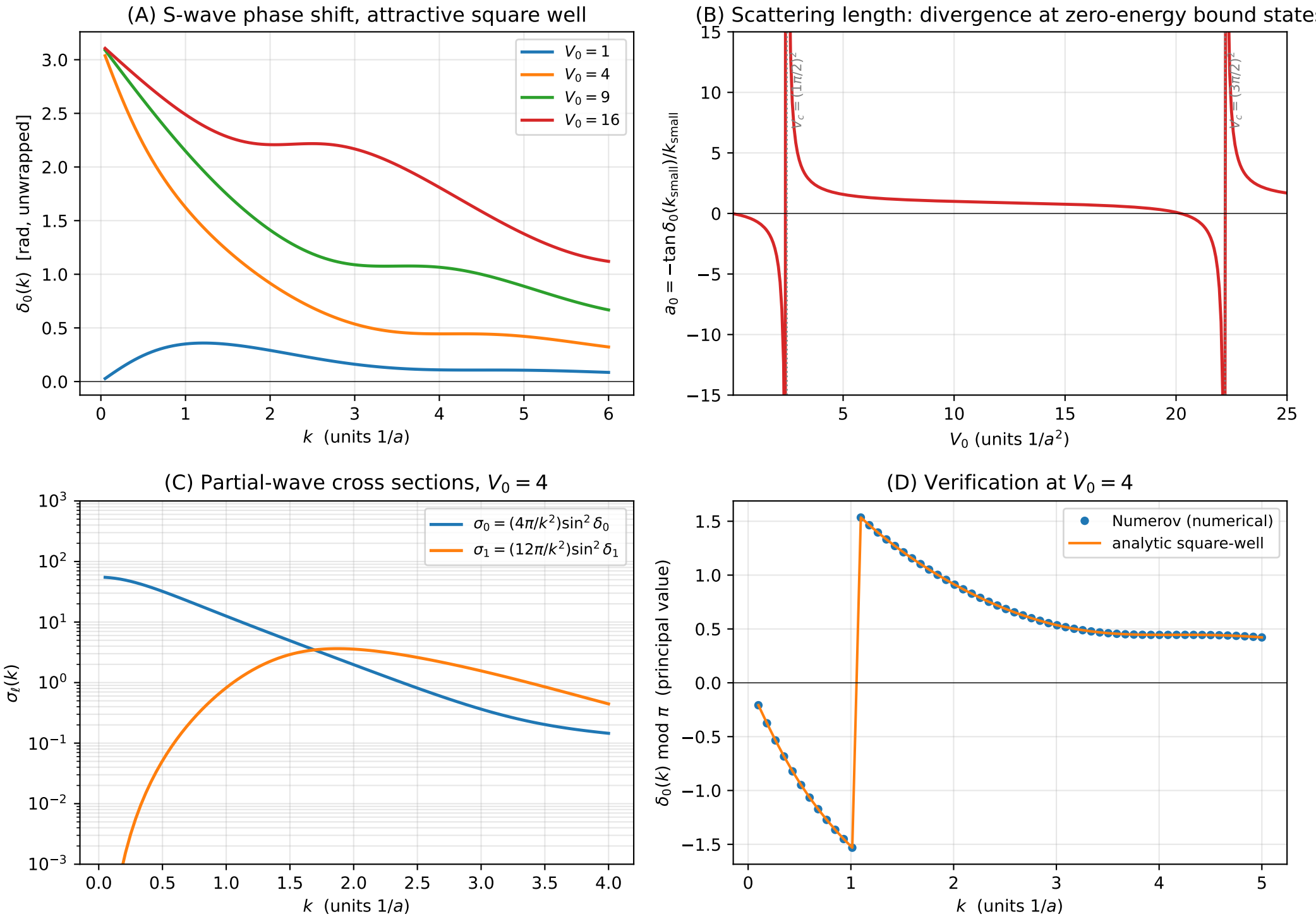
Numerov phase shifts: attractive square well $V(r) = -V_0\,\theta(a-r)$, units $\hbar = 2m = a = 1$
(A) S-wave phase shift, attractive square well
$V_0 = 1$
$V_0 = 4$
$V_0 = 9$
$V_0 = 16$
$\delta_0(k)$ [rad, unwrapped]
k (units $1/a$)
(B) Scattering length: divergence at zero-energy bound states
$a_0 = -\tan\delta_0(k_{\rm small})/k_{\rm small}$
V_0 (units $1/a^2$)
$V_c = (1\pi/2)^2$
$V_c = (3\pi/2)^2$
(C) Partial-wave cross sections, $V_0 = 4$
$\sigma_0 = (4\pi/k^2)\sin^2\delta_0$
$\sigma_1 = (12\pi/k^2)\sin^2\delta_1$
$\sigma_\ell(k)$
k (units $1/a$)
(D) Verification at $V_0 = 4$
Numerov (numerical)
analytic square-well
$\delta_0(k)$ mod π (principal value)
k (units $1/a$)